Karthik Bharadwaj

karthikbharadwajds@gmail.com | +1 (617) 238-8138 | Boston, MA | linkedin.com/in/karth1kb
| github.com/karthik-b-2001

Summary

Backend-focused software engineer with 2 years of experience building RESTful APIs, data pipelines, and cloud-deployed services using Python, FastAPI, PostgreSQL, and AWS. Pursuing an M.S. in Computer Science at Northeastern University (4.0 GPA) with coursework in ML, NLP, and AI. Eager to contribute to scalable, AI-driven backend systems in a fast-paced startup environment.

Experience

Digitide Solutions Ltd **Sep 2024 – Jun 2025**
Digital Associate – Software Engineer Bangalore, India

- Developed an NLP-powered chatbot that integrated AI models with backend services, enabling warehouse staff to query inventory via natural language and reducing manual lookup time by 60%.
- Delivered RESTful API-driven SaaS modules for AB-InBev serving 500+ daily users, including report generation, admin dashboards, and bulk data workflows.
- Architected and deployed backend services on Azure using Docker containers, reducing release turnaround time by 2 days.

zevvo **Jan 2024 – Jul 2024**
Software Engineer – Full Stack Bangalore, India

- Built scalable backend systems from scratch for bookings, billing, and catalog management with MongoDB schema design, JWT authentication, and RESTful API conventions, onboarding 50+ active users within the first quarter.
- Integrated third-party payment and notification SDKs into the backend service layer, enabling real-time transaction processing and reducing checkout drop-off by 15%.
- Streamlined CI/CD via GitHub Actions, deploying Dockerized services to AWS (ECS, S3, ECR) with auto-scaling and increasing deploy frequency by 40%.

Crestron Electronics **Mar 2023 – Jul 2023**
Software Development Intern Bangalore, India

- Optimized .NET Core REST APIs with rewritten SQL queries, cutting average response time by 28% across production endpoints.
- Elevated unit test coverage from 65% to 89% through Jest suites, reducing post-deployment bugs and shortening the QA feedback loop.

Education

Northeastern University **Expected Dec 2027**
Master of Science in Computer Science; GPA: 4.0/4.0 Boston, MA

- Coursework: Algorithms, DBMS, Foundations of AI, Programming Design Paradigms, Machine Learning, NLP

R V College of Engineering **May 2023**
Bachelor of Engineering in Computer Science and Engineering; CGPA: 8.04/10 Bangalore, India

Skills

- **Languages:** Python, SQL, Java, TypeScript, JavaScript, R, C++, C#
- **Backend:** FastAPI, Node.js, .NET Core, Express.js, SQLAlchemy, REST APIs, Microservices, Event-Driven Architecture
- **Frontend:** React.js, Next.js, Angular, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Firebase
- **Cloud/DevOps:** AWS (S3, ECS, Lambda, EC2), GCP, Azure, Docker, Kubernetes, GitHub Actions, CI/CD
- **ML/AI:** PyTorch, scikit-learn, OpenCV, YOLOv3, CNNs, Transfer Learning, NLP, AI Model Integration
- **Practices:** API Design, System Design, Distributed Systems, Caching, OOP, Agile, TDD, Git

Projects

ParetoFolio – Portfolio Optimization Engine Python, FastAPI, Pydantic v2, React 19, TypeScript

- Constructed a FastAPI backend with Pydantic v2 request validation, yfinance data ingestion, and a covariance matrix computation pipeline, serving a typed RESTful API consumed by a React 19 frontend.
- Implemented NSGA-II entirely from scratch in Python, including SBX crossover, polynomial mutation, tournament selection, and crowding distance, removing dependency on external optimization libraries.
- Designed a two-objective optimization strategy for portfolio construction, producing a ranked set of Pareto-optimal portfolios across 75 US equities with an interactive efficient frontier plot and Sharpe-ranked table.

SportsTV Streaming Data Warehouse R, MySQL 8, SQLite, R Markdown, Kimball Star Schema

- Engineered a bulk-batched, idempotent ETL data pipeline in R with surrogate-key resolution and partition pruning, achieving a 99%+ source-to-warehouse match rate across 2M+ streaming transactions.
- Modeled a Kimball-style star schema with five conformed dimensions and a RANGE-partitioned fact table on cloud-hosted MySQL 8, consolidating heterogeneous data sources into a unified analytical warehouse.
- Generated a fully automated BI report with KPI summaries, growth trends, and device breakdowns, driven entirely by live SQL queries with zero hard-coded values.

Autonomous Vehicle Perception Pipeline Python, PyTorch, YOLOv3, OpenCV, NumPy

- Fine-tuned YOLOv3 on a custom traffic sign dataset, achieving 88% mAP for real-time detection across varying lighting and weather conditions.
- Created a multi-sensor fusion pipeline combining camera and distance inputs for autonomous steering and braking, enabling lane-keeping without manual waypoint programming.